

MAXIMILIAN LIPSKI

Rendering Programmer | C++, Vulkan | Real-Time Rendering

+49 15751824418 max.lipski@hotmail.com maxlipski.dev linkedin.com/in/maximilian-lipski/ Dresden, Germany

SUMMARY

Master's student with half a year of experience in machine learning research and real-time rendering, experience in C++ and Vulkan. Key achievements include developing a real-time rendering engine achieving around 400 FPS at 2560x1600 resolution and building an RHI with Vulkan support and future backend expansions.

EXPERIENCE

Student Research Assistant

TU Dresden - ScaDS.AI

11/2025 - 04/2026 Dresden, DE

http://scads.ai

- Researched machine learning approaches to accelerate geometric rendering of geometric objects
- Developed and trained neural networks for geometric approximation problems using CUDA-enabled machine learning workflows
- Investigated neural representations of Hermite spline tubes as signed distance functions
- Trained a neural network for ray-intersection with Hermite spline tubes
- Designed and evaluated experiments to analyze the accuracy and performance of learned geometric approximations
- Achieved promising results for fixed primitives while identifying parameterization as a major challenge or generalization

PROJECTS

Rodan - Real-Time Rendering Engine

- Created a real-time rendering engine built on top of a custom Rendering-Hardware Interface (RHI) with a Vulkan backend
- Achieves around 400 FPS while rendering the Sponza scene at 2560x1600 on an RTX 5070
- Built a physically based rendering (PBR) pipeline with image-based lighting (IBL)
- Developed an HDR tone mapping pipeline
- Added glTF 2.0 asset loading with support for KHR_materials_transmission and KHR_materials_volume
- Integrated Dear ImGui for in-engine tooling and debugging
- Debugged and profiled rendering workloads using NVIDIA Nsight Graphics and RenderDoc

Velos - RHI

- Designed and implemented a backend-independent RHI with a Vulkan backend
- Developed abstractions for pipelines, descriptor sets, textures, samplers, buffers, and command lists
- Built automatic SPIR-V shader reflection for descriptor set and pipeline layout generation
- Architected the system to support future graphics backends such as DirectX 12

EDUCATION

Master of Science, Computer Science Technical University Dresden

Dresden, DE

- Specialization: Computer Graphics
- Expected Graduation: 01/2027

Bachelor of Science, Computer Science Heinrich-Heine University

10/2020 - 09/2024 Düsseldorf, DE

- Grade: 1.9
- Thesis: Development of a Skew-Heap Data Structure for Persistent Memory

SKILLS

Programming

C/C++ GLSL CUDA

Graphics APIs

Vulkan OpenGL

Tools

RenderDoc NVIDIA Nsight Graphics
NVIDIA Nsight Systems Git CMake
Premake Unreal Engine 5

Platforms

Linux Windows

LANGUAGES

German Native ●●●●●
Polish Native ●●●●●
English C1 ●●●●●